

# Submission D solution to Problem 1

We consider infinite countable first-order structures, their automorphism groups and the descriptive aspects of their representations. A representation  $\mathcal{A}$  of such a structure is an instance of its isomorphism type in which the universe or domain of the structure is the set of natural numbers  $\mathbb{N}$  and the constants, functions and relations of the structure are elements of  $\mathbb{N}$ , multi-variable functions from  $\mathbb{N}$  to  $\mathbb{N}$  and multi-place relations on  $\mathbb{N}$ , respectively. Similarly, an automorphism of  $\mathcal{A}$  or an isomorphism from  $\mathcal{A}$  to  $\mathcal{B}$  is a bijective function from  $\mathbb{N}$  to  $\mathbb{N}$  that preserves the evaluation of all constants, functions and relations. We let  $\text{AUT}(\mathcal{A})$  denote the automorphism group of  $\mathcal{A}$ .

**Definition 1.** •  $\mathcal{A}$  is computably AUT-countable if every automorphism of  $\mathcal{A}$  is computable relative to  $\mathcal{A}$ .

- $\mathcal{A}$  is computably AUT-countable on a cone if there is a  $C \subset \mathbb{N}$  such that whenever  $\mathcal{B}$  is isomorphic to  $\mathcal{A}$ , every automorphism of  $\mathcal{B}$  is computable relative to  $\mathcal{B} \oplus C$ .

**Definition 2.** A function  $f : \mathcal{A} \rightarrow \mathcal{A}$  is  $\Sigma_1^{in}$ -definable in  $\mathcal{A}$  relative to parameters  $\bar{a} = (a_1, \dots, a_n)$  from  $\mathcal{A}$  if there is a family of existential formulas  $(\phi_i)_{i \in \mathbb{N}}$  such that for all  $x$  and  $y$  in  $\mathcal{A}$ ,  $f(x) = y$  if and only if there is an  $i \in \mathbb{N}$  such that  $\mathcal{A} \models \phi_i(\bar{a}, x, y)$ .

**Question 1.** Is there a countable computably-represented structure  $\mathcal{A}$  such that  $\mathcal{A}$  is computably AUT-countable on a cone and such that for any finite set of parameters  $\bar{a}$  in  $\mathcal{A}$  there is an automorphism  $\pi$  of  $\mathcal{A}$  such that  $\pi$  is not  $\Sigma_1^{in}$ -definable in  $\mathcal{A}$  relative to the parameters  $\bar{a} \cup \pi(\bar{a})$ , where  $\pi(\bar{a})$  is the image of  $\bar{a}$  under  $\pi$ .

## Solution

*Proof.* We will prove that the answer to the question is negative: there is no such structure. We proceed by contradiction. Assume that there exists a countable computably-represented structure  $\mathcal{A}$  satisfying the two conditions in the problem.

**Step 1: The automorphism group of  $\mathcal{A}$  is countable.**

By the first condition,  $\mathcal{A}$  is computably AUT-countable on a cone. Thus, there exists an oracle  $C \subset \mathbb{N}$  such that for every structure  $\mathcal{B}$  isomorphic to  $\mathcal{A}$ , every automorphism of  $\mathcal{B}$  is computable relative to  $\mathcal{B} \oplus C$ . Applying this to  $\mathcal{B} = \mathcal{A}$ , we have that every automorphism of  $\mathcal{A}$  is computable relative to the oracle  $\mathcal{A} \oplus C$ . Because there are only countably many Turing machines, there are only countably many functions that are computable relative to any fixed oracle. Therefore, the set of all automorphisms of  $\mathcal{A}$  is countable, which means the group  $\text{AUT}(\mathcal{A})$  is countable.

**Step 2: The group  $\text{AUT}(\mathcal{A})$  contains an isolated point.**

The group  $\text{AUT}(\mathcal{A})$  is a closed subgroup of the symmetric group  $S(\mathbb{N})$  equipped with the topology of pointwise convergence. Therefore,  $\text{AUT}(\mathcal{A})$  is a Polish space (a separable and completely metrizable topological space). By the Baire Category Theorem, any non-empty countable Polish space must contain at least one isolated point. Because  $\text{AUT}(\mathcal{A})$  is a topological group, its topology is homogeneous, meaning that if one point is isolated, then all points are isolated. In particular, the identity element  $\text{id} \in \text{AUT}(\mathcal{A})$  is an isolated point.

**Step 3: There exists a finite tuple with a trivial pointwise stabilizer.**

In the topology of pointwise convergence on  $\text{AUT}(\mathcal{A})$ , the basic open neighborhoods of the identity  $\text{id}$  are the pointwise stabilizers of finite tuples of elements from  $\mathbb{N}$ . Because  $\text{id}$  is an isolated point, the singleton set  $\{\text{id}\}$  is open. Thus, there exists a basic open neighborhood of  $\text{id}$  that is exactly  $\{\text{id}\}$ . This implies there exists a finite tuple  $\bar{c}$  in  $\mathcal{A}$  such that if  $\sigma \in \text{AUT}(\mathcal{A})$  and  $\sigma(c_i) = c_i$  for all elements  $c_i$  in  $\bar{c}$ , then  $\sigma = \text{id}$ .

**Step 4: Fixing the parameters and the automorphism  $\pi$ .**

We choose our finite set of parameters to be the tuple  $\bar{a} = \bar{c}$ . By the second condition of the problem, there exists an automorphism  $\pi$  of  $\mathcal{A}$  such that  $\pi$  is not  $\Sigma_1^{in}$ -definable in  $\mathcal{A}$  relative to the parameters  $\bar{c} \cup \pi(\bar{c})$ . Let  $\bar{d} = \pi(\bar{c})$ . We define the expanded structure  $\mathcal{A}^* = (\mathcal{A}, \bar{c}, \bar{d})$ , which is the structure  $\mathcal{A}$  together with new constant symbols evaluating to the elements

of  $\bar{c}$  and  $\bar{d}$ .

**Step 5: Rigidity of the expanded structure  $\mathcal{A}^*$ .**

Let  $\sigma$  be an automorphism of the expanded structure  $\mathcal{A}^*$ . By definition,  $\sigma$  is an automorphism of  $\mathcal{A}$  that fixes the interpretations of the new constants, meaning  $\sigma(\bar{c}) = \bar{c}$  and  $\sigma(\bar{d}) = \bar{d}$ . Because  $\sigma(\bar{c}) = \bar{c}$ , Step 3 implies that  $\sigma = \text{id}$ . Therefore, the only automorphism of  $\mathcal{A}^*$  is the identity. Consequently, every relation on the domain of  $\mathcal{A}^*$  is invariant under all automorphisms of  $\mathcal{A}^*$ . In particular, the binary relation  $R = \{(x, y) \in \mathbb{N}^2 : \pi(x) = y\}$  (the graph of  $\pi$ ) is invariant under all automorphisms of  $\mathcal{A}^*$ .

**Step 6: Application of the Ash–Knight–Manasse–Slaman and Chisholm Theorem.**

We rely on the following theorem from computable structure theory:

**Theorem** (Ash–Knight–Manasse–Slaman and Chisholm). *Let  $M$  be a countable first-order structure, and let  $R$  be a relation on the domain of  $M$ . Suppose that  $R$  is invariant under all automorphisms of  $M$ . Suppose there exists an oracle  $C \subset \mathbb{N}$  such that for every structure  $N$  with domain  $\mathbb{N}$  and every isomorphism  $f : M \rightarrow N$ , the relation  $f(R)$  is computable relative to  $N \oplus C$ . Then  $R$  is definable in  $M$  by an  $L_{\omega_1\omega}$   $\Sigma_1$  formula without parameters (meaning there is a countable family of parameter-free existential formulas in the language of  $M$  whose disjunction defines  $R$ ).*

We verify the hypotheses of this theorem for  $M = \mathcal{A}^*$  and  $R = \text{graph}(\pi)$ :

1.  $M = \mathcal{A}^*$  is a countable first-order structure, as  $\mathcal{A}$  has domain  $\mathbb{N}$  and we added finitely many constants.
2.  $R$  is a relation on the domain of  $M$ .
3.  $R$  is invariant under all automorphisms of  $M$ , which was established in Step 5.
4. Let  $C$  be the fixed oracle from Step 1. Let  $N$  be any structure with domain  $\mathbb{N}$  and let  $f : \mathcal{A}^* \rightarrow N$  be an isomorphism. The structure  $N$  is an expansion of a structure  $\mathcal{B}$  by constants  $f(\bar{c})$  and  $f(\bar{d})$ , where  $\mathcal{B}$  is a structure in the language of  $\mathcal{A}$ . The map  $f$  is an isomorphism from  $\mathcal{A}$  to  $\mathcal{B}$ .

The relation  $f(R)$  is  $\{(f(x), f(y)) : \pi(x) = y\}$ , which is exactly the graph of the function  $f \circ \pi \circ f^{-1}$ . Because  $\pi$  is an automorphism of  $\mathcal{A}$

and  $f : \mathcal{A} \rightarrow \mathcal{B}$  is an isomorphism,  $f \circ \pi \circ f^{-1}$  is an automorphism of  $\mathcal{B}$ .

By the definition of  $C$ , every automorphism of  $\mathcal{B}$  is computable relative to  $\mathcal{B} \oplus C$ . Therefore,  $f \circ \pi \circ f^{-1}$  is computable relative to  $\mathcal{B} \oplus C$ . Because the atomic diagram of  $N$  (viewed as a real in  $2^\omega$ ) consists of the atomic diagram of  $\mathcal{B}$  and the evaluations of the finitely many constants, the oracle  $\mathcal{B} \oplus C$  is Turing equivalent to  $N \oplus C$ . Thus,  $f(R)$  is computable relative to  $N \oplus C$ . This verifies the fourth hypothesis.

**Step 7: Reaching a contradiction.**

By the theorem,  $R$  is definable in  $\mathcal{A}^*$  by a countable disjunction of parameter-free existential formulas. This means there is a family of existential formulas  $(\phi_i(x, y))_{i \in \mathbb{N}}$  in the language of  $\mathcal{A}^*$  such that for all  $x$  and  $y$  in  $\mathbb{N}$ ,  $R(x, y)$  holds if and only if there is an  $i \in \mathbb{N}$  such that  $\mathcal{A}^* \models \phi_i(x, y)$ .

The language of  $\mathcal{A}^*$  is the language of  $\mathcal{A}$  expanded by constant symbols for  $\bar{c}$  and  $\bar{d}$ . By replacing these constant symbols with variables, we obtain a family of existential formulas  $(\psi_i(\bar{u}, \bar{v}, x, y))_{i \in \mathbb{N}}$  in the language of  $\mathcal{A}$  such that for all  $x$  and  $y$  in  $\mathbb{N}$ ,  $\pi(x) = y$  if and only if there is an  $i \in \mathbb{N}$  such that  $\mathcal{A} \models \psi_i(\bar{c}, \bar{d}, x, y)$ .

This matches the exact definition given in the problem statement for the function  $\pi$  being  $\Sigma_1^{in}$ -definable in  $\mathcal{A}$  relative to the parameters  $\bar{c} \cup \bar{d}$ . Because  $\bar{d} = \pi(\bar{c})$ , we conclude that  $\pi$  is  $\Sigma_1^{in}$ -definable in  $\mathcal{A}$  relative to the parameters  $\bar{c} \cup \pi(\bar{c})$ . This directly contradicts the initial choice of  $\pi$  from Step 4.

**Conclusion:**

Because the assumption that such a structure  $\mathcal{A}$  exists leads to a logical contradiction, we conclude that no such countable computably-represented structure exists. The answer to Question 1 is no.  $\square$